ELIXIR-SI tools & services

ELIXIR All Hands 2022, 7-10 June, Amsterdam, Netherlands

CENTRAL TOOLS & SERVICES - CORE FACILITIES

ELIXIR-SI HPC national data node

1. **National Archive of Life Science Research Data / Central repository**
   - for data from various fields of life sciences (including e.g. fEGA)
   - in collaboration with other complementary (ESFRI) RIs
2. **Core network equipment and Compute cluster (CPU and GPU)**
   - HPC consisting of 256-core CPU nodes and NVIDIA A100 80G GPU nodes
   - Connected to national HPC via Geant 100Gbps
3. **ELIXIR-SI eLearning Platform (EeLP)**
   - https://elixir.mf.uni-lj.si

LABORATORIES

Central Next Generation Sequencing laboratory
- NextSeq2000 (Illumina)
- nCounter® SPRINT Profiler
- SeqStudio™ Genetic Analyzer
- QuantStudio 7 Pro with Real-Time PCR module and Studio3D digital PCR module
- Confocal microscope Leica SP8
- RIEGL miniVUX-2UAV
- Nadia Innovative (Dolomite Bio)
- MARVIN ProLine seed analyser

Laboratory for single cell analysis
- Confocal Microscope ZEISS LSM 900
- 10x Genomics Chromium Controller
- Gridion Nanopore

ELIXIR-SI RI-SI-2 research equipment

https://elixir-slovenia.org/elixir-si-activities/

IJS
- Network switch for access to the central data hub
- Server for temporary local storage and local data analysis

KI
- Network switch for access to the central data hub
- Server for temporary local storage and local data analysis
- FlowJo software
- Server with high-performance GPU cards

KIS
- Network switch for access to the central data hub
- Server for temporary local storage and local data analysis

NIB
- Network switch for access to the central data hub
- Server for temporary local storage and local data analysis

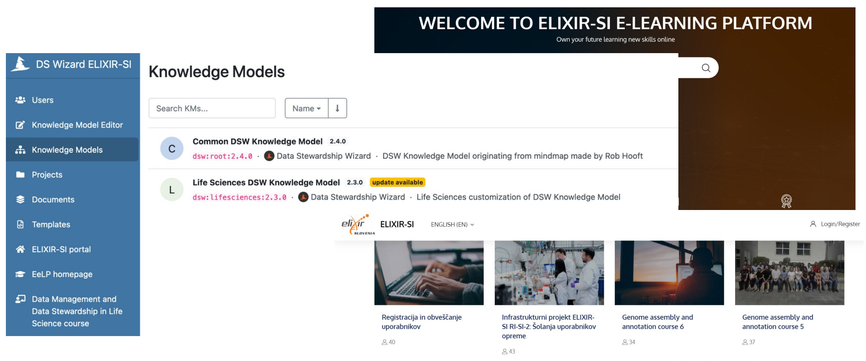
UL BF
- Network switch for access to the central data hub
- Server for temporary local storage and local data analysis

UL FRI
- Network switch for access to the central data hub
- Server for temporary local storage and local data analysis

UL VF
- Network switch for access to the central data hub
- Server for temporary local storage and local data analysis

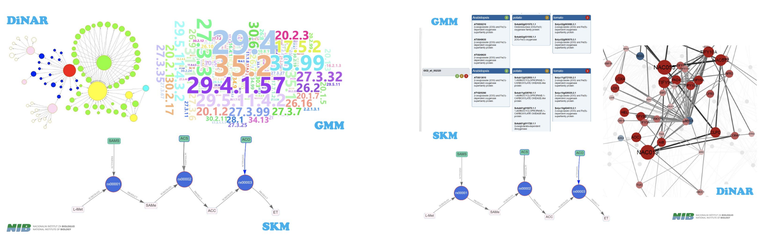
UL MF (LJ) and UL VF (MB)
- National Data Node

UM
- Network switch for access to the central data hub
- Server for temporary local storage and local data analysis



single entry point

- **website:** https://elixir-slovenia.org
- **informative order of service:** https://elixir-slovenia.org/order-a-service/



DiNAR (Differential Network Analysis in R) - user friendly application for revealing hidden patterns of plant signalling dynamics achieved by using dynamic visualisation of complex experiments and by identification of strong differential interactions that reveal hidden patterns. Visualisations can be extended to any user defined network in combination with experimental (omics) datasets - https://nib-si.shinyapps.io/DiNAR/ (NIB)
GMM (GoMapMan) is an open web-based resource for gene functional annotations in plant sciences that covers protein coding genes, metabolites and small RNAs placed in the plant specific MapMan ontology, with assistance of phylogenetic relationships derived from various orthology clusterings - https://gomapman.nib.si/ (NIB)
SKM (StressKnowledgeMap) - web application that enables the standardised construction of an integrated knowledge network of the plant stress response. SKM merges existing knowledge on model species and potato molecular physiology into a large knowledge network. https://skm.nib.si/ (NIB)

ELIXIR-SI members TOOLS & SERVICES

UL FRI offers (1) **analysis of transcriptomics data** using genome-scale metabolic models in a combination with different model extraction methods; (2) **analysis of time series experimental data** to identify rhythmic measurements as well as to characterise their rhythmicity; (3) **inference of causal networks** (e.g., Boolean representation of gene regulatory networks) from time series data (e.g., representing gene expression data). Integration of these approaches can be used to represent a comprehensive biological system (systems biology).

In development (UL MF)
- **Galaxy service** (on ELIXIR-SI HPC national data node)
- **fEGA national node** ((on ELIXIR-SI HPC data node)
- **Bioinformatics** core services by staff scientists

ELIXIR-SI eLearning Platform (EeLP - https://elixir.mf.uni-lj.si**)** - hosts all training courses, materials and other educational resources from ELIXIR-SI and other ELIXIR nodes. Open educational resources are linked with TeSS. EeLP is part of ELIXIR Training Platform - https://elixir-europe.org/platforms/training (UL MF)

ELIXIR-SI Data Stewardship Wizard - DSW instance for developing specific knowledge model used by Slovenian life science research community and for training of Slovenian life science researchers - https://elixir-si.ds-wizard.org/ (UL MF)

ELIXIR-SI compute (CPU and GPU) standard SLURM HPC cluster. AAI access for ELIXIR community is in development in collaboration with other ELIXIR nodes. (UL MF)

Workflows (including development and acceleration of existing algorithms) **for processing and visualizing** large amounts of data obtained with the **cryo-electron microscope**. (KI)

Online tools for prediction of life-science related properties of molecules, such as protein transmembrane segments predictor (https://tinyurl.com/d01-trp) or classification of compounds interacting with P-glycoprotein (https://www.vegahub.eu/). (KI)

NGS sequencing - at core facilities and ELIXIR-SI members facilities on Illumina, Nanopore and Ion Proton technologies. Short term plan is to upgrade facilities with automated systems for sample preparation (and loading). (UL MF, UL BF, UL VF, UKCLJ, KI)

Commercial software: CLC Genomics Server (UL BF) and FlowJo (UL MF, KI)

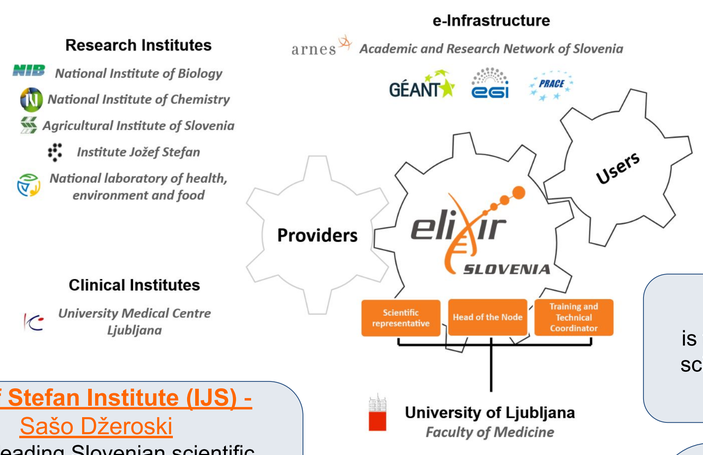
ELIXIR-SI PARTNERS

National Institute of Chemistry (KI) - Marjana Novič
is the second largest natural sciences research institute in Slovenia. Research at KI is divided into two major fields: life and materials science.

National Institute of Biology (NIB) - Kristina Gruden
is the largest independent Public Research Institution for Life Sciences in Slovenia. NIB's diverse research undertakings revolve around solutions to environmental, agricultural, food and human health issues.

National Laboratory of Health, Environment and Food (NLZOH) - Maja Rupnik
is the central and largest Slovenian public health laboratory that handles environmental protection, diagnostic and public health microbiological activities, chemical and microbiological analyses of different types of samples, and also performs research activities.

Agricultural Institute of Slovenia (KIS) - Barbara Gerič Stare and Jelka Šuštar Vozlič
is the leading research institute in the field of agriculture in Slovenia. It comprehensively deals with the issues of modern agriculture and is expanding its activities into the fields of environmental protection and ecology.



Jožef Stefan Institute (IJS) - Sašo Džeroski
is the leading Slovenian scientific research institute, covering a broad spectrum of basic and applied research in natural sciences, life sciences and engineering.

University Medical Centre Ljubljana (UKCLJ) - Jernej Kovač
is the largest tertiary health care institution in Slovenia and one of important drivers of biomedical research in the region. UKCLJ pursues health care, education and research and is committed to providing safe, high quality and standardised patient care.

University of Ljubljana (UL)
is the oldest and largest higher education and scientific research institution in Slovenia. Four faculties are members of ELIXIR-SI.

Faculty of Medicine, UL MF - coordinator of ELIXIR-SI
is the leading educational and research faculty in medicine in Slovenia. Coordinates and runs central tools and services of ELIXIR-SI that support three main goals: data node (national data archive and compute nodes), wet-lab (generate experimental laboratory data) and training node (traning events and EeLP - https://elixir.mf.uni-lj.si).

Computer and Information Science, UL FRI - Miha Mraz
Focused to the development of computational pipelines for the analysis and integration of omics as well as other types of experimental data.

Biotechnical Faculty, UL BF - Jernej Jakše
Generate and analyse genomics and transcriptomics data.

Veterinary Faculty, UL VF - Marko Cotman
Generate and analyse different experimental data from veterinary samples (mainly from companion animals and livestock).

ARNES - Avgust Jauk in Tomi Dolenc
is a public institute that provides network services to research, educational and cultural organizations, and enables them to establish connections and cooperation with each other and with related organizations abroad.

University of Maribor (UM) - Urban Bren
is the 2nd largest and 2nd oldest university in Slovenia. It represents an autonomous research and education facility which fosters development of new knowledge and its transfer into environment in the areas of natural, social, technical and medical sciences. Faculty of Chemistry and Chemical Engineering from UM is partner in ELIXIR-SI.

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